

Temporal Consistency in Human–Object Interaction: Why the Spoon Usually Moves Before the Mug

Alicia García, Robert Smith and Jane Doe

Laboratory for Everyday Video Analysis, Fictional Institute of Applied Scenes

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Abstract—We present a small temporal model for checking the order of object interactions in short videos. The model combines hand proximity, object motion, and persistence into event proposals, then penalizes implausible orderings. In a synthetic tabletop benchmark it reduces isolated false events, while occasionally suppressing valid unusual actions. The results argue for explicit uncertainty around commonsense temporal rules.

Keywords—video analysis; temporal reasoning; reproducibility; synthetic experiments.

1 Introduction

Video systems can recognize objects yet still tell an odd story about them. A mug may appear to move before the hand reaches it because detection is delayed. We explore a lightweight temporal check for spoon and mug interactions, using a familiar sequence as an example rather than a universal law.

2 Related Work

Prior fictional work models tabletop interactions as event graphs [1] and smooths short detection gaps [2]. A cautionary study of uncommon routines shows why a strong prior should never become a hard rule [3]. Our method keeps the prior soft.

3 Methodology

A frame-level detector supplies hand, spoon, and mug positions. Motion above a threshold begins a candidate event, and three consecutive quiet frames end it. We score a sequence by the sum of detector log scores minus λ for an unsupported order inversion. The parameter $\lambda = 0.7$ was selected on a separate synthetic development set.

4 Experimental Setup

The benchmark contains 90 rendered clips of 12 seconds at 25 frames/s. Sixty show spoon-first interaction, 20 mug-first interaction, and ten contain no contact. Camera angle and frame dropouts vary. Event start and end times are known exactly from the renderer.

5 Event Representation

We define contact as a hand-object distance below a synthetic threshold for at least two frames. An onset is recorded at the first supporting frame, while an offset requires three quiet frames. These asymmetric conditions prevent a single missed detection from fragmenting an otherwise continuous manipulation. The model keeps a null event when evidence is insufficient; it does not force every clip to contain a spoon or mug action.

Each candidate path can include a short gap, a complete event, or no event. Dynamic programming evaluates these paths in timestamp order. The temporal prior is applied only after the detector evidence is accumulated, so a clear mug-first sequence can overcome the usual spoon-first preference. We store the best and second-best paths. Their score difference is a simple diagnostic for clips where the reported order depends heavily on the prior.

6 Validation Design

We divide clips by scene seed, reserving 30 for development and 60 for evaluation. The evaluation set contains both common and intentionally reversed action orders. An event counts as correct when its object label agrees and its onset is within 120 ms of the renderer timestamp. We report event F1 and the number of spurious motions; sequence accuracy alone would hide errors in event boundaries. All simulated frame losses are applied after ground-truth events are generated.

7 Results

Temporal checking raised event F1 from 0.79 to 0.85 on the full set. It reduced false spoon movements during brief occlusions from 14 to six. In the 20 valid mug-first clips, however, two correct sequences were incorrectly reordered. This error is consequential because the unusual cases may be of greatest interest.

Method	Event F1	False events
Frame cues	0.79	14
Temporal check	0.85	6

Synthetic benchmark summary.

8 Discussion

An ordering prior can repair noisy evidence without proving how a task ought to be performed. The detector must expose alternatives and confidence, particularly for clinical or educational observations where atypical order is informative. Synthetic timing also understates real annotation uncertainty.

9 Sensitivity to the Prior

At $\lambda = 0$, the system is equivalent to detector-only decoding. A modest penalty removes several short, unsupported spoon events. When the penalty exceeds 1.4, the model begins to reorder clear mug-first clips, despite strong visual evidence. We therefore treat the selected value as a property of this benchmark, not a parameter ready for deployment. A distribution of plausible sequences would be more informative than a single forced explanation.

10 Error Review

The remaining false events often occur when a hand passes near the spoon without touching it. Motion alone is then ambiguous: the spoon may be occluded, or it may remain stationary. Three clips also contain nearly simultaneous object movement, for which an exact order is not physically meaningful at the available frame rate. Reporting both event intervals and their uncertainty avoids claiming a distinction below temporal resolution.

A mug-first action is neither anomalous nor incorrect in the general case. The synthetic dataset includes it precisely to test whether the model preserves uncommon but valid evidence. In the two incorrectly reordered examples, the

detector confidence dips for four frames during hand occlusion. Replacing the gap with ideal detections restores the correct order, suggesting that uncertainty propagation is the more promising remedy than a stronger rule.

11 Limits of Interpretation

A detected event is an observation about motion and proximity, not a judgment about intent. The same temporal pattern may arise while stirring, setting a table, or clearing it. Camera angle can hide contact, and a person's hand can move one object while occluding another. An external activity label would be needed to interpret the sequence as a particular task. These limitations are especially important when an unusual order is treated as informative: the system should present the underlying frames and timing instead of silently correcting the observation.

12 Reporting Recommendation

For each clip, we would retain the detector proposals, the selected event sequence, and the runner-up sequence with their scores. A reviewer could then inspect whether a change in the temporal prior alters the substantive finding. Aggregate F1 should be accompanied by a breakdown for common order, reversed order, and no-contact clips. This compact audit trail is more useful than presenting a confident sequence with no indication of competing interpretations.

13 Conclusions and Future Work

A soft temporal constraint improves event coherence in this controlled example. Future work will estimate order priors from diverse users and report alternative event sequences when evidence is weak.

References

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